

Figure 12.7 Transactions from account 0x2812c... on Ropsten public chain

12.8.2 A solution

How do you achieve confidentiality in blockchain applications? To protect the confidentiality of data, a novel concept called zero-knowledge-proof has been proposed. Zcash (https://z.cash/technology) is a working solution for realizing confidentiality in cryptocurrency transfers. It achieves confidentiality by implementing a novel type of Txs called shielded transactions or z Txs. In this realm, unshielded Txs are referred to as t Txs. Zcash is well-founded in a strong scientific background. It offers four types of Txs:

- Both sender (z) and receiver (z) are shielded (fully private).
- Only sender (z) is shielded, not the receiver (t).
- Only receiver (z) is shielded, not the sender (t).
- Both receiver and sender are not shielded (public).

In this case, z refers to shielding or hiding, and t refers to a regular nonshielded entity. Although this solution is available